

WGAN-based Data Reconstruction Attack (GLASS)

Anonymous CVPR submission

Paper ID *****

Abstract

As deep learning becomes increasingly pervasive, concerns about privacy have grown, particularly in collaborative cloud-edge environments. In this work, we rebuild **GLASS** (GAN-based Latent Space Search) [2], a unique GAN-based method for Data Reconstruction Attacks (DRA) targeting Split Inference (SI) systems. Conventional DRA techniques have trouble efficiently utilizing public data and frequently fall short at deeper split points where little information is available. Instead of using StyleGAN for reconstruction, we built it using Wasserstein GAN with Gradient Penalty (WGAN-GP), which is known for stable training and high-fidelity outputs. **GLASS** leverages the latent space of pre-trained WGAN-GP models to overcome these difficulties and produce more accurate reconstructions. We show through extensive experiments on CIFAR-10 that our approach outperforms current DRA techniques by a significant margin, even when tested against advanced defense systems such as DISCO [3]. Our findings highlight critical vulnerabilities in SI systems and demonstrate that **GLASS** provides a strong remedy for SI privacy issues by being highly successful at recreating private inference data.

1. Introduction

Concerns about sensitive data privacy have grown in importance as deep learning models are used more and more for important applications. One promising method for addressing edge devices' computational constraints and safeguarding user privacy in cooperative cloud-edge configurations is Split Inference (SI) [1]. To ensure that only intermediate data representations—rather than the raw input—are transferred, SI splits the model, with a portion of it operating on the edge device and the remainder on the cloud. Despite its benefits, SI is still susceptible to Data Reconstruction Attacks (DRA), in which adversaries try to use the intermediate representations to reassemble private input data.

The majority of the DRA techniques now used in SI are based on learning and optimization strategies. By de-

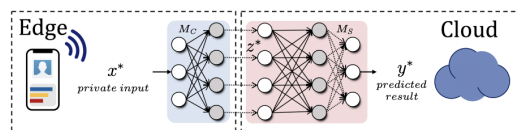


Figure 1. Split Inference

creasing the Euclidean distance between intermediate representations, optimization-based assaults aim to identify the closest input data; nevertheless, they frequently struggle with deep splits and are unable to make good use of public data. Because learning-based approaches depend on auxiliary data and need a high level of similarity between the public dataset and the inference data, they are extremely vulnerable to changes in the distribution of data.

In order to get over these restrictions, we present **GLASS**, a GAN-based attack technique that makes use of pre-trained WGAN with Gradient Penalty models and their latent space to recreate data more successfully. A thorough assessment of **GLASS** against a range of split points and defensive mechanisms is one of our contributions, demonstrating that our strategy outperforms existing DRA techniques in terms of efficacy and resilience.

2. Related Work

Reconstructing private inference data from the shared intermediate representations in Split Inference (SI) systems is the goal of Data Reconstruction Attacks (DRA). The possible privacy hazards that these attacks present in machine learning applications—particularly when handling sensitive user data—have drawn attention. DRAs can be divided into two main categories: optimization-based and learning-based techniques.

1. Optimization-based DRAs: DRA is treated as an optimization problem by early techniques like Likelihood Maximization (LM) and regularized Maximum Likelihood Estimation (rMLE). The aim is to minimize the discrepancy between the intermediate representation and the anticipated outcome from possible input samples. Although rMLE aligns the reconstructed image with the distribution

of natural images by using Total Variation (TV) regularization, it is unable to effectively utilize the publicly available data that could improve reconstruction accuracy. In the same way, LM enhances optimization-based assaults, but it is still susceptible to the caliber of auxiliary data, especially when the distribution of data is very different from the inference data.

2. **Learning-based DRAs:** These methods, such as the Inverse Network (IN) method, transfer intermediate feature representations back to the original input data using learning models. However, when the auxiliary data is limited or not representative of the target data, learning-based approaches perform worse and require large amounts of data.

3. **Defense mechanisms:** Alongside advancements in DRA, defense strategies have been proposed to harden SI systems. Methods such as channel obfuscation, feature compression, and noise injection aim to disrupt the reconstruction ability of adversaries. DISCO [3] dynamically identifies and obfuscates sensitive channels to degrade reconstruction quality without significantly impacting classification accuracy. However, such defenses often involve a delicate balance between privacy and performance, and sophisticated attacks like GLASS can still achieve substantial reconstruction quality under these defenses.

Even though these current methods have advanced significantly, they still have issues with scalability, flexibility to different kinds of data, and resilience to defenses. On the other hand, our study presents GLASS, which makes use of latent space search and pre-trained WGAN-GP models to greatly enhance attack performance by leveraging publicly available data.

3. Methodology

Overview:

Pre-trained WGAN-GP models are known for producing high-quality images, and our attack technique, GLASS (GAN-based Latent Space Search), expands on their strength. By identifying the ideal latent vector that produces intermediate feature representations that are comparable to those seen in Split Inference systems, we take advantage of the latent space of these GANs to reconstruct data.

The methodology consists of the following components:

1. **Latent Space Search:** In order to identify the closest match to an intermediate representation, traditional DRA techniques frequently rely on optimization. However, because the reconstruction problem is non-convex, this method may become trapped in local minima. In order to overcome this difficulty, GLASS looks for the optimal latent vector in WGAN-GP's **Z-space**. This reduces the possibility of local minima during the optimization process and increases the probability of obtaining a suitable solution. To keep the reconstruction realistic, the search procedure is



Figure 2. Image produced by DISCO and our model

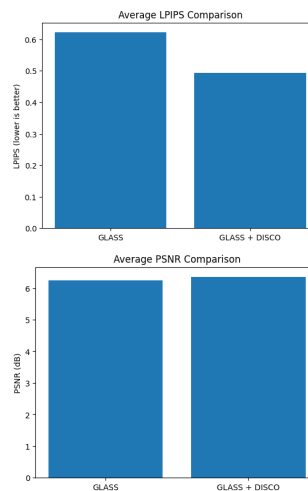


Figure 3. Comparisons of LPIPS and PSNR metrics of GLASS and DISCO

regularized using a **Kullback-Leibler (KL)** divergence to limit the latent vectors to follow a Gaussian distribution.

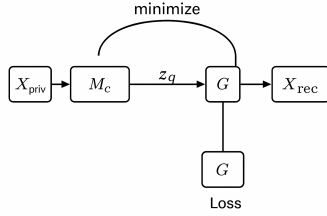


Figure 4. Attack Framework, 'Xpriv' is the private data, 'Xrec' is the reconstructed output of the model, 'G' depicts the Generator of WGAN-GP and 'Mc' represents the Client Model, z_q is the intermediate representation

2. Loss Function and Total Loss: We use two main loss functions in our assault strategy: the **reconstruction loss** and the **WGAN loss**. These losses are intended to preserve high-quality generating properties while guaranteeing that the rebuilt image is as similar to the original as possible.

- **Reconstruction Loss:** The difference between the original picture x and the reconstructed image $\hat{x}$ (obtained after generating from the latent space) is measured by the **reconstruction loss**:

$$\mathcal{L}_{recon} = \|x - \hat{x}\|_2^2 \quad (1)$$

The model is encouraged by this loss to reduce the pixel-by-pixel discrepancy between the original and reconstructed images.

- **WGAN Loss:** To guarantee that the produced images have a distribution similar to that of real photos, the **WGAN (Wasserstein GAN) loss** is applied. The discriminator D attempts to distinguish between produced and genuine images. The definition of the WGAN loss is:

$$\mathcal{L}_{WGAN} = \mathbb{E}[D(x)] - \mathbb{E}[D(\hat{x})] \quad (2)$$

In order to ensure high-quality picture generation that corresponds with actual data distributions, this loss incentivizes the generator to create images that can deceive the discriminator.

- **Total Loss:** The reconstruction and WGAN losses are combined into the overall loss function, which is weighted by the corresponding coefficients λ_1 and λ_2 :

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{recon} + \lambda_2 \mathcal{L}_{WGAN}$$

Achieving high reconstruction accuracy and superior generative results requires striking a balance between these two losses. The relevance of each component can be adjusted during training thanks to the coefficients λ_1 and λ_2 .

4. Experiments

Datasets and Tasks:

We evaluated GLASS:

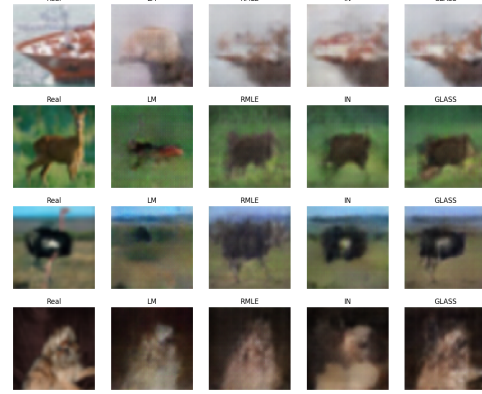


Figure 5. Comparison between Data Reconstruction Attacks, here we can observe that our model's reconstruction are better than other reconstructions.

CIFAR10: Contains 60,000 images with each class having 6,000 images. It has a large collection of inanimate and animate objects, with different backgrounds, shapes, and colors. This dataset was utilized to provide a baseline, which will help identify any problems with a particular type of data.

Attack Setup:

The attack consists of the following components. 1. To replicate split inference we constructed a CNN based cifar-10 classifier that takes in the image as an input and produces the label as an output. This was then split into the server and client. The client has two convolution layers and ReLU activation, this gives out the intermediate representation (z_q), while the server has a fully connected layer which produces the label when z_q is given as an input. This model was trained for 200 epochs and gives an accuracy of 64.93%. 2. Under **white-box attack** conditions, we ran tests in which the adversary only gets intermediate representations but has complete access to the model architecture. The **WGAN-GP** model was trained using the **private dataset (DP)** and the **auxiliary dataset (DA)**.

Evaluation Metrics:

We used **LPIPS**, **SSIM**, **PSNR**, and **MSE** to evaluate the quality of the reconstructed images.

5. Results

We evaluate GLASS against different defense mechanisms like Dynamic and Invariant Sensitive Channel Obfuscation (DISCO) [3] for deep neural networks, as well as against existing Data Reconstruction Attack (DRA) techniques like Regularized Maximum Likelihood Estimation

(rLME), Likelihood Maximization (LM), and Inverse Networks (IN). The goal of the tests is to determine how effective these assaults are in various adversary configurations and split points.

The comparison between the different DRA attacks is shown in 5, which clearly shows that when compared with optimization-based DRA's our model works the best. In addition, the only attack that is similar or better is Inverse Networks, which is a learning-based DRA attack.

When DISCO defense is applied to the images it removes the perturbations and provides a clean RGB image as indicated in 2. This is helpful as we can then calculate the scoring metrics between this and all the other attacks. Which will give pure results on how well the model works when defenses are set.

The 3 provides the scores for both when the image has been reconstructed by glass and when DISCO is used to remove the perturbation. It depicts that the reconstructed image is very close to the original as the LPIPS score between them is not big.

GLASS perform exceptionally well, particularly at deeper split points where conventional DRA techniques fall short. The analysis shows that reconstruction stability is greatly enhanced with **GLASS++**, particularly in difficult situations with little intermediate data.

6. Conclusion

The reconstruction accuracy of intermediate representations from the split inference model using **WGAN-GP** is the best among all the other Data Reconstruction attacks. While other methods came close they still lacked in image quality. While other attacks use different metrics to reconstruct the data, our GAN-based model makes use of latent space search. In the future, we will investigate and work to recreate **GLASS++** and expand its use to include different datasets and attack scenarios. We would also like to experiment with dis-continious GAN models like **VQ-GAN** to produce a working Data Reconstruction Attack.

Project link:[here](#)

References

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